

SAN ANGELO, TX, USA

WMO: 722630

Lat: 31.352N

Lon: 100.495W

Elev: 584

StdP: 94.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23034

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -3.5	(d) -14.4	(e) 1.2	(f) 3.4	(g) -12.0	(h) 1.4	(i) 4.7	(j) 12.8	(k) 16.6	(l) 11.6	(m) 13.4	(n) 3.3	(o) 30	(p) 0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.1	(c) 39.0	(d) 21.3	(e) 37.8	(f) 21.2	(g) 36.6	(h) 21.2	(i) 24.1	(j) 32.3	(k) 23.6	(l) 31.7	(m) 23.1	(n) 31.1	(o) 4.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 22.1	(b) 18.0	(c) 26.7	(d) 21.5	(e) 17.4	(f) 26.3	(g) 21.0	(h) 16.8	(i) 26.0	(j) 75.5	(k) 32.1	(l) 73.2	(m) 31.5	(n) 71.5	(o) 31.2	(p) 26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.4	8.5	-9.1	40.9	2.0	1.6	-10.5	42.0	-11.7	42.9	-12.8	43.8	-14.2	45.0
			-10.1	25.0	1.9	0.9	-11.5	25.7	-12.6	26.2	-13.7	26.7	-15.1	27.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	8.6	10.8	15.0	19.5	24.1	27.7	29.0	28.8	24.9	19.4	13.4	9.0
	DBStd	8.60	5.29	5.75	5.40	4.72	4.06	2.83	2.31	2.53	3.45	4.50	5.06	5.08
	HDD10.0	271	90	54	18	2	0	0	0	0	0	2	24	81
	HDD18.3	1188	303	215	128	42	6	0	0	0	2	42	159	290
	CDD10.0	3642	46	77	174	287	437	531	590	582	447	294	127	51
	CDD18.3	1517	1	4	26	77	184	281	332	323	200	76	11	1
	CDH23.3	18428	34	101	319	939	2099	3453	4327	4087	2141	773	124	31
	CDH26.7	9699	4	26	96	387	1053	1898	2512	2359	1058	280	22	4
Wind	WSAvg	4.2	4.3	4.5	4.7	5.0	4.6	4.4	3.8	3.5	3.4	3.7	4.0	4.1
Precipitation	PrecAvg	530	23	29	30	40	76	58	28	56	76	63	29	22
	PrecMax	909	93	115	118	130	286	178	118	206	279	289	132	101
	PrecMin	249	0	0	0	0	3	1	0	0	0	0	0	0
	PrecStd	160	22	27	26	34	57	43	26	47	58	55	29	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	30.0	32.0	35.9	38.7	40.6	40.1	40.4	38.4	34.0	29.3	26.7
		MCWB	13.0	14.0	15.6	17.2	19.6	21.5	21.3	21.7	20.7	19.8	16.9	13.8
	2%	DB	23.6	26.3	29.4	33.2	36.4	38.3	38.5	38.7	35.9	31.6	26.6	23.5
		MCWB	12.1	13.7	15.5	17.3	19.8	21.2	21.3	21.8	20.9	19.7	16.2	13.4
	5%	DB	20.8	23.7	26.9	30.9	34.6	36.6	37.3	37.4	34.1	29.9	24.5	20.8
		MCWB	11.5	13.0	15.1	17.3	19.6	21.3	21.4	21.6	20.8	18.8	15.7	12.3
	10%	DB	18.2	20.9	24.5	28.7	32.4	34.9	36.2	35.9	32.5	27.9	22.3	18.3
		MCWB	10.6	12.2	14.6	16.8	19.9	21.6	21.5	21.5	20.7	18.2	14.9	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	17.9	19.2	21.5	24.0	25.1	24.7	24.4	24.0	22.9	20.1	17.8
		MCDB	20.7	23.1	25.6	28.7	32.0	33.3	33.1	33.1	31.3	28.8	24.6	21.4
	2%	WB	15.0	16.7	18.3	20.5	23.0	24.2	23.8	23.7	23.1	21.8	18.7	15.8
		MCDB	19.5	21.7	24.2	27.2	30.3	31.9	32.3	32.5	30.0	27.7	22.8	19.7
	5%	WB	13.2	15.3	17.4	19.8	22.3	23.5	23.2	23.2	22.5	20.9	17.5	14.0
		MCDB	18.4	20.3	23.1	26.1	29.4	31.3	31.7	31.8	29.4	26.5	22.1	18.9
	10%	WB	11.4	13.7	16.4	18.9	21.6	22.9	22.7	22.7	21.9	20.0	16.2	12.1
		MCDB	17.3	19.6	22.1	25.2	28.7	30.7	31.2	31.2	28.6	25.1	20.9	17.4
Mean Daily Temperature Range	5% DB	MDBR	14.4	14.4	14.4	15.0	13.5	12.8	13.1	12.9	12.9	13.7	13.7	14.1
		MCDBR	18.6	18.5	17.1	17.3	16.2	15.0	14.5	14.5	15.0	15.8	16.0	17.9
		MCWBR	9.9	9.6	8.0	7.5	5.8	4.4	4.1	4.2	5.0	6.5	8.1	9.5
	5% WB	MCDBR	14.6	14.6	13.3	13.5	13.2	12.5	12.6	12.7	12.0	11.6	11.9	13.6
		MCWBR	9.1	9.4	7.1	6.8	5.5	4.4	3.9	3.9	4.4	5.5	7.3	8.6
Clear-Sky Solar Irradiance	taub		0.286	0.294	0.317	0.354	0.374	0.390	0.401	0.389	0.373	0.336	0.308	0.291
	taud		2.564	2.545	2.492	2.399	2.406	2.400	2.425	2.453	2.426	2.522	2.529	2.552
	Ebn at Noon		953	974	967	936	913	893	882	892	897	913	916	928
	Edh at Noon		82	92	104	119	119	119	116	111	110	93	84	79
All-Sky Solar Radiation	RadAvg		3.30	4.03	5.19	6.27	6.52	7.13	6.95	6.41	5.33	4.33	3.47	2.96
	RadStd		0.35	0.52	0.39	0.47	0.57	0.49	0.47	0.38	0.39	0.62	0.42	0.34

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	+87
(47) Regional (19 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+100

Nomenclature: See separate page